

Guo Chen

Software Engineer

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EDUCATION

Singapore Institute of Technology & DigiPen Institute of Technology Aug 2022 - Apr 2026

Bachelor of Science with Honours in Computer Science in Real-Time Interactive Simulation

Nanyang Polytechnic

Mar 2017 - Feb 2020

Diploma in Nanotechnology & Materials Science

PROJECTS

AI Chat Platform

Feb 2026 - Recent

React | Java Spring Boot | PostgreSQL | AWS S3 | Redis

- Developed a full-stack interactive platform using React and Spring Boot, incorporating a dynamic model selection feature and a dedicated admin management dashboard.
- Architected a PostgreSQL and AWS S3 storage solution, integrating a Redis caching layer to optimize query performance and securely manage distributed user sessions, chat history, and multi-media uploads with metadata persistence.

Pawsie Adventure

Aug 2024 - Mar 2025

3D Puzzle Platformer game | C | C++ | OpenGL | ImGui | Visual Studio

- Led the development of rendering algorithms and UI components using ECS architecture, ensuring high-efficiency resource management.
- Optimized 3D graphics modules to improve system performance and user interaction within a complex simulation environment.

Feline Felony

Aug 2023 - Mar 2024

2D Puzzle Platformer game | C | C++ | OpenGL | ImGui | Visual Studio

- Designed and implemented optimized 2D graphics modules using OpenGL, enhancing real-time rendering efficiency.
- Delivered a visually immersive experience through high-performance interactive simulation systems.

EXPERIENCE

Software Engineering(intern)

May 2025 - Apr 2026

ST Engineering

Developing a modular simulation component to procedurally generate entities (buildings, vehicles, characters) on virtual terrain for CNN training. Responsible for:

- Generating diverse, realistic datasets for AI model training.
- Building a configurable UI and UAV simulation for synthetic image capture.
- Supporting system architecture and user manual documentation.

Project focuses on simulation-based data generation for computer vision applications.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, JavaScript, Python
- **Graphics & Rendering:** OpenGL, Vulkan
- **Frontend Development:** React, JavaScript, HTML, CSS, Qt
- **Backend Development:** Spring Boot, RESTful APIs, AWS S3, Redis
- **Mobile Development:** Android, Jetpack Compose
- **Database:** MySQL, PostgreSQL